

2(a)	correctly stated formula ; e.g. actual diameter = image length / magnification magnification = image length / actual diameter allow a magnification triangle 1750 (μm) ; <i>(for 70 mm) credit also measurements made to 0.5 mm</i> A 1700 (μm) <i>(for 68 mm)</i> A 1725 (μm) <i>(for 69 mm)</i> A 1775 (μm) <i>(for 71 mm)</i> A 1800 (μm) <i>(for 72 mm)</i>	2
2(b)(i)	squamous epithelial (tissue) ; A pavement epithelial (tissue)	1

Question	Answer	Marks
2(b)(ii)	<i>any two from:</i> (alveoli have) thin wall / wall is one cell thick ; A wall composed of thin cells R thin membrane R thin cell wall <i>idea of</i> short diffusion distance ; <i>in context of between alveolar space and, blood / capillary</i> elastic tissue, qualified ; A elastin / elastic fibres e.g. stretch and recoil for, ventilation / inhalation and exhalation maintaining steep, diffusion / partial pressure, gradient (many) alveoli provide a large surface area (for diffusion) ; I <i>idea that a single alveolus has a large surface area</i> AVP ; surfactant, prevents alveolar collapse / maintains inflation I <i>ref. to</i> moist lining	2
2(c)	<i>any three from:</i> - loosely packed spongy mesophyll cells / spongy mesophyll cells surround air spaces / air spaces between spongy mesophyll cells ; - large, surface / area, for evaporation (of water to air space) ; <i>In context of spongy mesophyll cells, surfaces / cell walls</i> (cells arranged so that), air spaces above the stomata / sub-stomatal air spaces ; <i>ref. to</i> evaporation creates saturated air spaces / water vapour collects in air spaces ; <i>idea of</i> creates water potential gradient between air in air spaces and air in external environment ; - AVP ; e.g. guard cells situated in epidermis to form stomatal pore <i>ref. to</i> xylem, close proximity to spongy mesophyll cells for passage of water / passes on water to spongy mesophyll cells	3

Question	Answer	Marks
2(d)(i)	<i>accept au for arbitrary units and first hour etc. for 08.00 etc.</i> <i>any three from:</i> 1 between 08.00 and 09.00, steep increase / increase from 2.5 au to 5 au ; 2 highest rate / peak, at 09.00 ; A maximum rate is at 09.00 3 fluctuates / AW, between 10.00 and 15.00 ; 4 (continued) decrease between 15.00 and 19.00 ; A steep /steepest, decrease between 15.00 and 16.00 A less steep (than 15.00 to 16.00) decrease between 16.00 and 19.00 5 comparative data quote for decrease ; e.g. 3.8 au at 15.00 to 0.5 au at 19.00 or 3.8 au at 15.00 to 2.00 au at 16.00 or 2.0 au at 16.00 to 0.5 au at 19.00	3
2(d)(ii)	<i>any two from:</i> <i>change in external environmental factor</i> decrease in light intensity / AW ; A (less sunlight because it was) cloudy decrease in temperature / AW ; <i>temperature decreased because it was cloudy is mp2 only</i> decrease in, wind speed / air movement / AW ; increase in humidity / raining / AW ; stomata, decrease in (aperture) size / open less wide ; decreased evaporation (rate) ; water vapour diffuses out less rapidly / less steep water potential gradient ; AW AVP ; e.g. suggestion of xerophytic feature occurring in response to increased, temperature / light intensity (to 09.00)	2